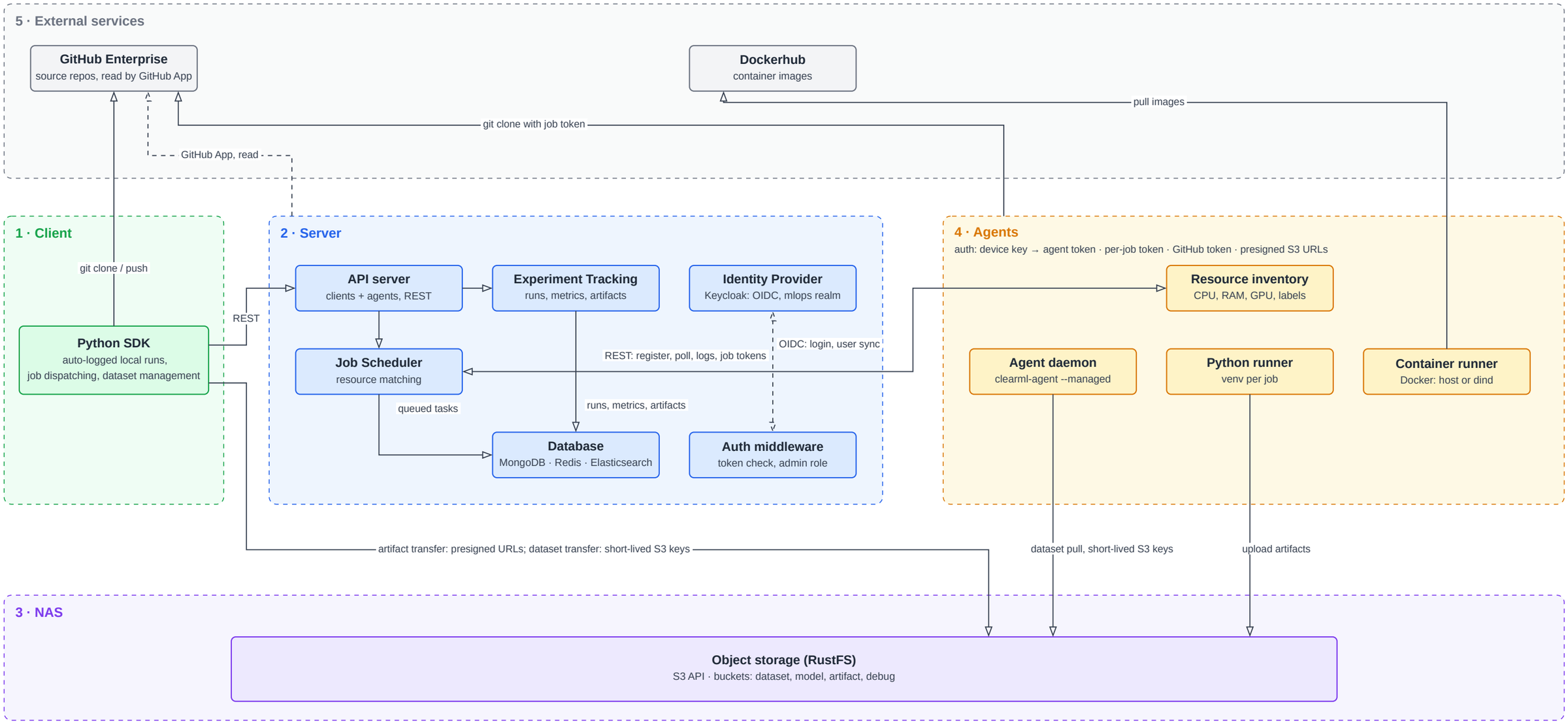
MLOps Framework: System Architecture

Five physical machines. Datasets and artifacts live in object storage; the server stores their metadata and references, never the files in its databases.



Machine rules

- **Client** is the Python SDK. It talks to the Server (REST), to GitHub Enterprise (git) and to the object store on the NAS (artifacts through presigned S3 URLs; datasets through DVC, with short-lived keys from the server) and keeps a per-user cache of datasets.
- **Server** runs the API, the scheduler, the database (MongoDB for tasks and metadata, Redis and Elasticsearch for caches and events) and the sign-in service (Keycloak), in the same deploy stack.
- **NAS** runs the object store: one S3-compatible service (RustFS) with four buckets (dataset, model, artifact, debug), and one service user that may only use those buckets. The server uses it to issue short-lived keys limited to the dataset bucket.
- **Agents** are enrolled by an admin with a device key. Each job gets a token, and a short-lived GitHub token that clones its repository from GitHub Enterprise. Jobs run in containers (Docker on the host, or a Docker daemon in a `dind` container). They pull images from Dockerhub, move artifacts with presigned S3 URLs and datasets with short-lived S3 keys, and keep a dataset cache that persists between jobs.

Data and identity

- Artifacts are ClearML artifacts. Datasets are GitHub repositories with their files in DVC: the commit is the version, and a run stores the repository, the commit and a hash of the files, not the files. Outside datasets (Hugging Face and others) are read through adapters kept in an external-datasets repository.
- Source repositories are on GitHub Enterprise. The server reads them with a read-only GitHub App; a job clones its repository with its own token.
- Clients enroll once through a Keycloak login, then use their device key only. Agents are enrolled by an admin only.
- Presigned S3 URLs name the object store’s public address, so clients must be able to reach it.
- Dashed arrows: identity and authentication checks.
- Users sign in through Keycloak, and the server syncs users from it.